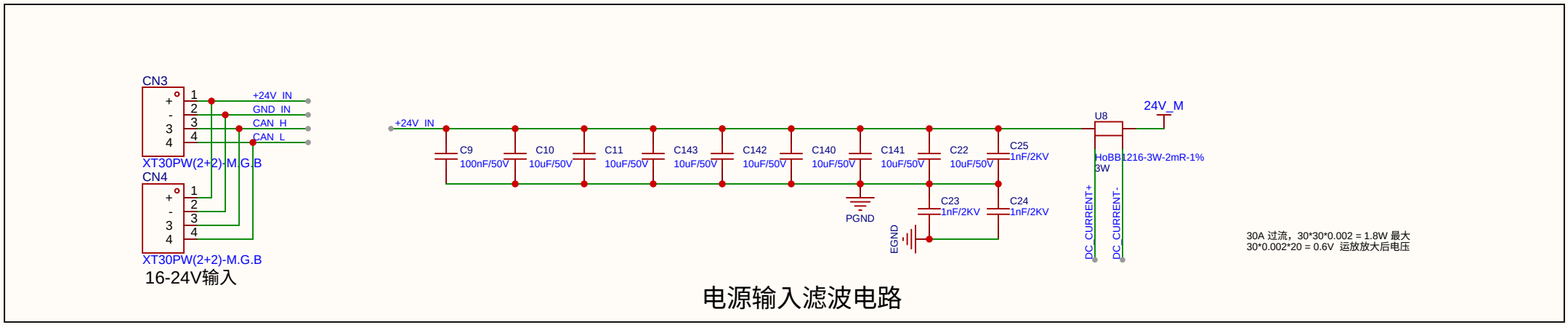
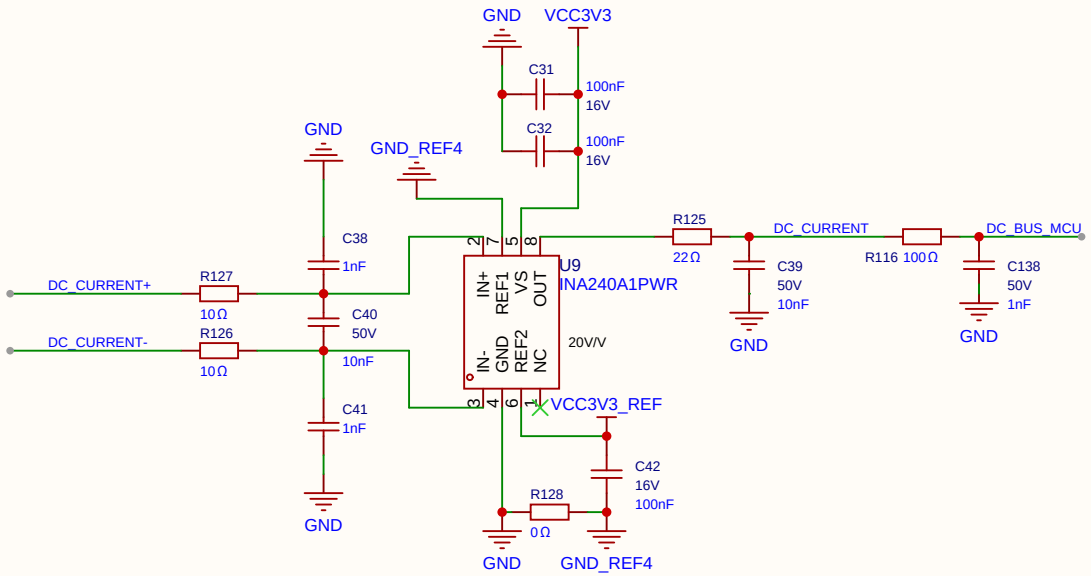
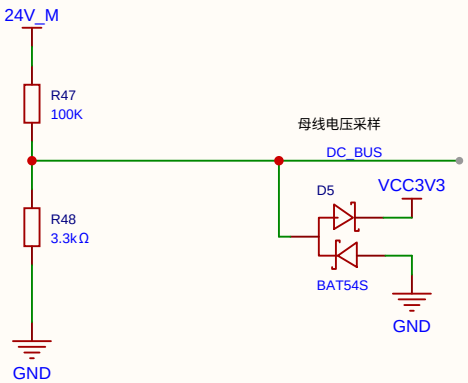


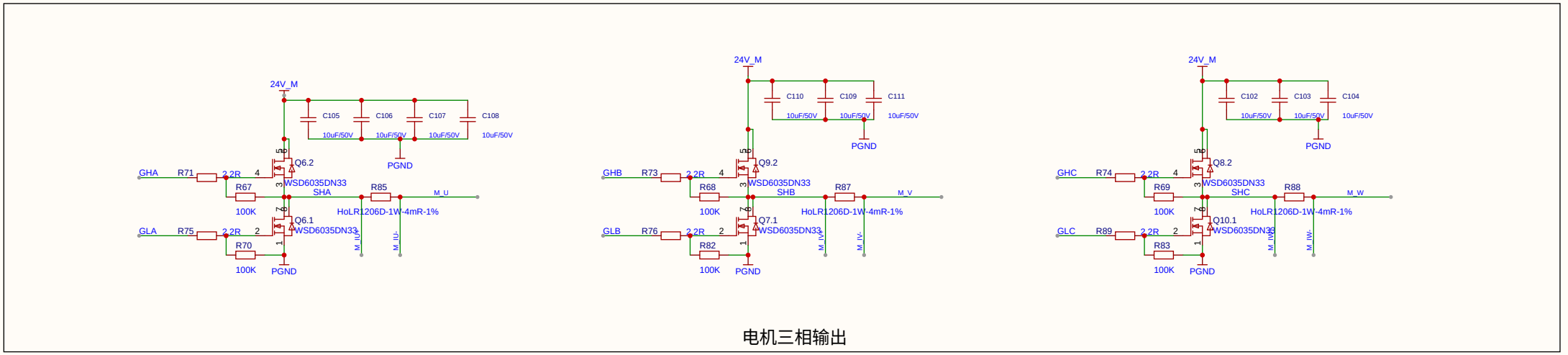
电源输入滤波电路



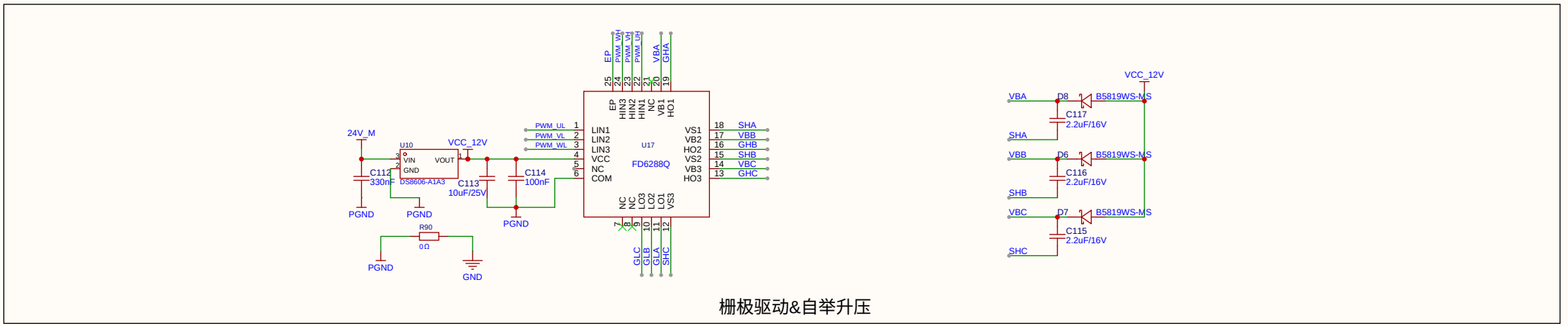
母线电流采样



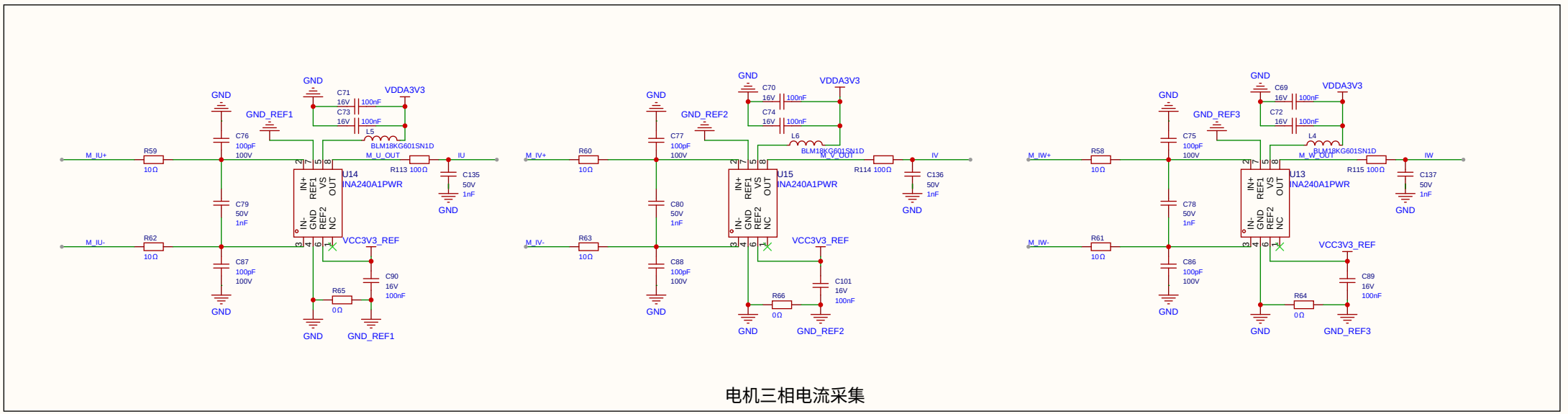
母线电压采样



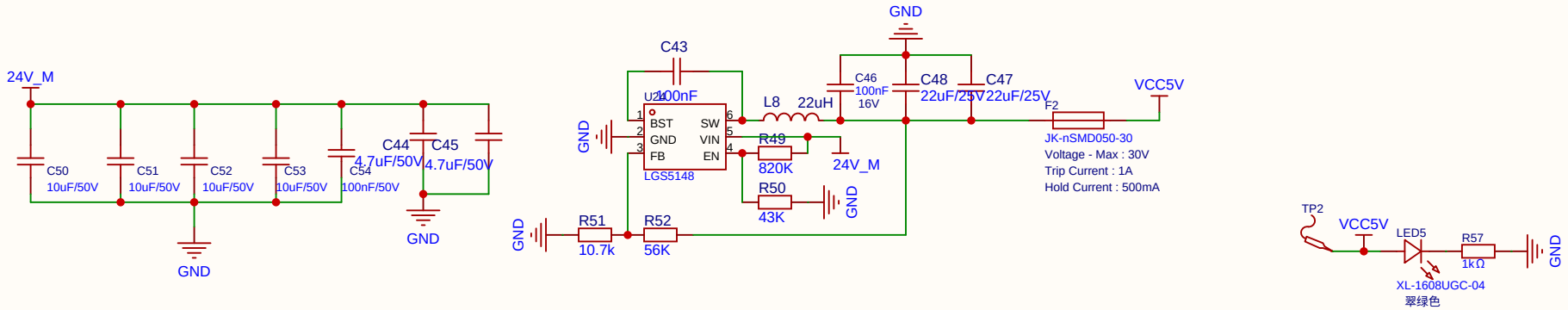
电机三相输出



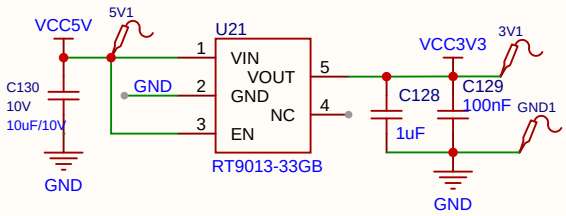
栅极驱动&自举升压



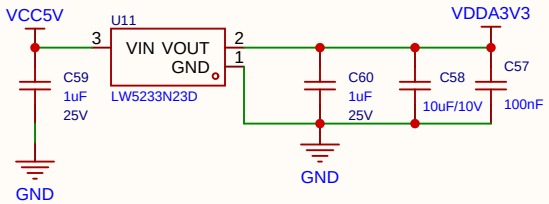
电机三相电流采集



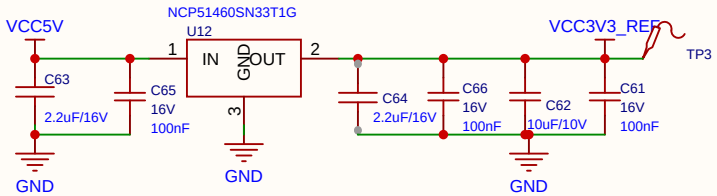
24-36V转5V输出&电源指示灯



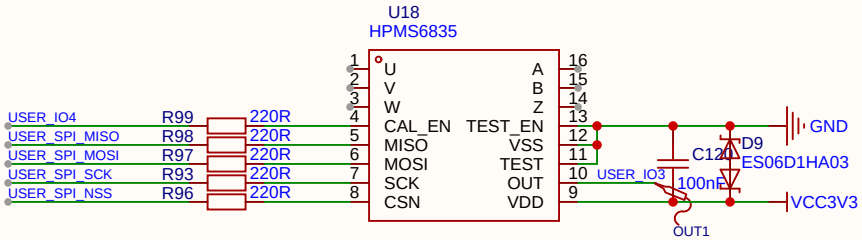
5V转3.3V数字域供电



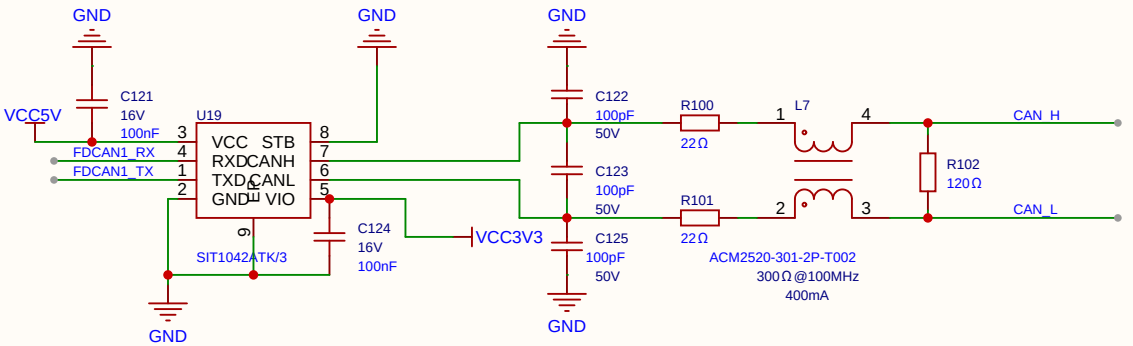
5V转3.3V模拟域供电



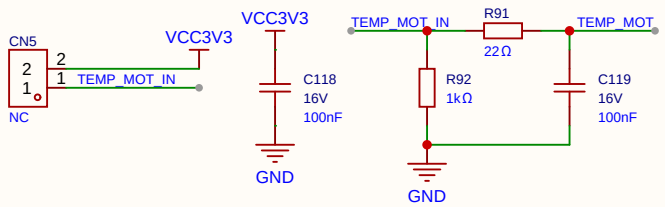
5V转3.3V参考电压



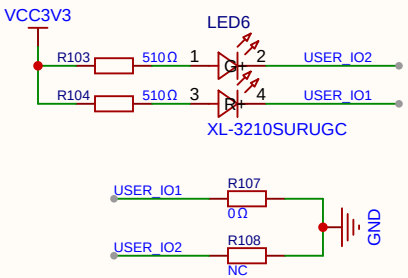
磁编码器



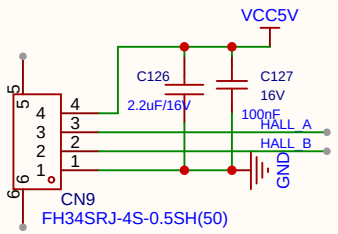
CANFD收发芯片



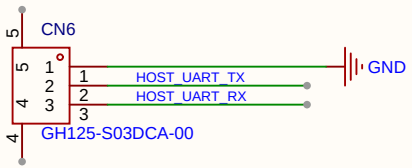
电机温度传感器接口



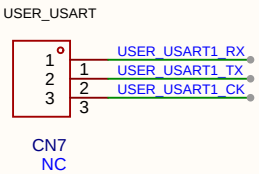
电机状态指示灯



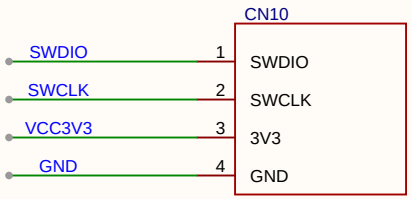
霍尔接口



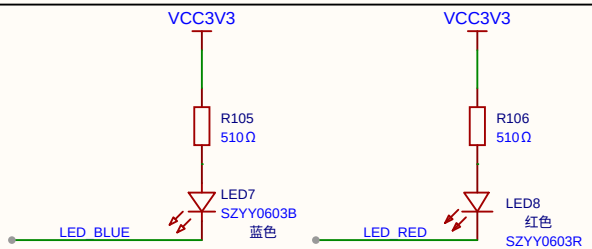
用户串口接口



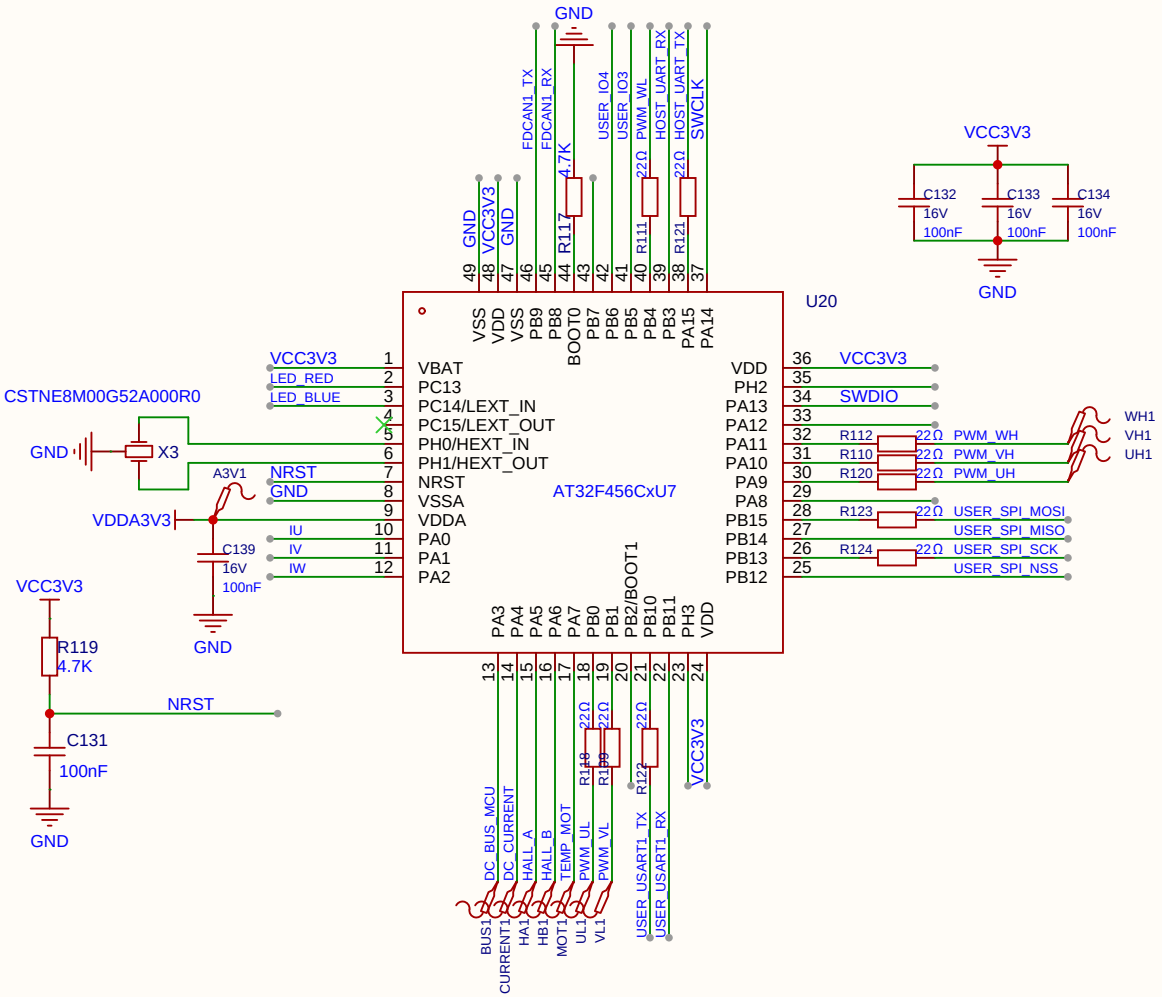
串口调试测试点



下载接口 SWD



电机调试指示灯


GND
VCC3V3
C132
16V
100nF
C133
16V
100nF
C134
16V
100nF
GND
U20
AT32F456CxU7
VBAT
PC13
PC14/LEXT_IN
PC15/LEXT_OUT
PH0/HEXT_IN
PH1/HEXT_OUT
NRST
VSSA
VDDA
PA0
PA1
PA2
VDD
PH2
PA13
PA12
PA11
PA10
PA9
PA8
PB15
PB14
PB13
PB12
VSS
VDD
VSS
PB9
PB8
BOOT0
PB7
PB6
PB5
PB4
PB3
PA15
PA14
PA3
PA4
PA5
PA6
PA7
PB0
PB1
PB2/BOOT1
PB10
PB11
PH3
VDD
VCC3V3
LED_RED
LED_BLUE
NRST
GND
IU
IV
IW
SWDIO
PWM_WH
PWM_VH
PWM_UH
USER_SPI_MOSI
USER_SPI_MISO
USER_SPI_SCK
USER_SPI_NSS
FDCAN1_TX
FDCAN1_RX
USER_IO4
USER_IO3
PWM_WL
HOST_UART_RX
HOST_UART_TX
SWCLK
R117
4.7K
R111
22Ω
R121
22Ω
R112
22Ω
R110
22Ω
R120
22Ω
R123
22Ω
R124
22Ω
R118
22Ω
R119
22Ω
R122
22Ω
WH1
VH1
UH1
CSTNE8M00G52A000R0
X3
A3V1
VDDA3V3
C139
16V
100nF
R119
4.7K
C131
100nF
DC_BUS_MCU
DC_CURRENT
HALL_A
HALL_B
TEMP_MOT
PWM_UL
PWM_VL
USER_USART1_TX
USER_USART1_RX
BUS1
CURRENT1
HA1
HB1
MOT1
UL1
VL1

MCU